WiL update, 22 October 2026

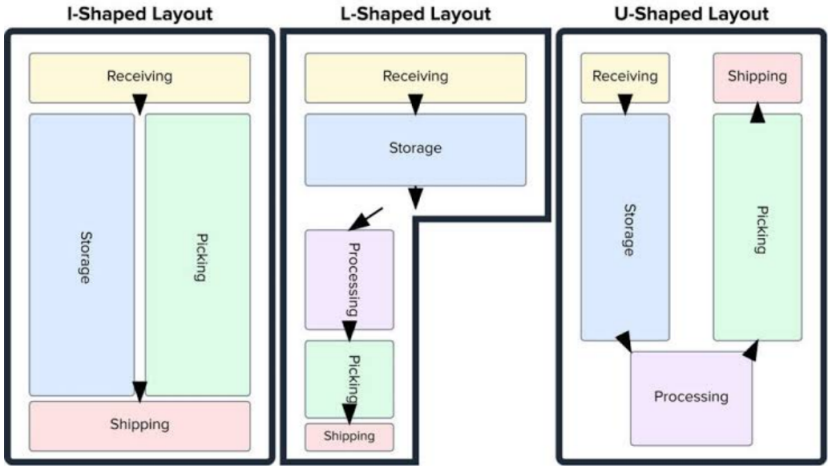
Realistic world · Experiment insights · Motion gate

Bolton Athit Davies

Contents

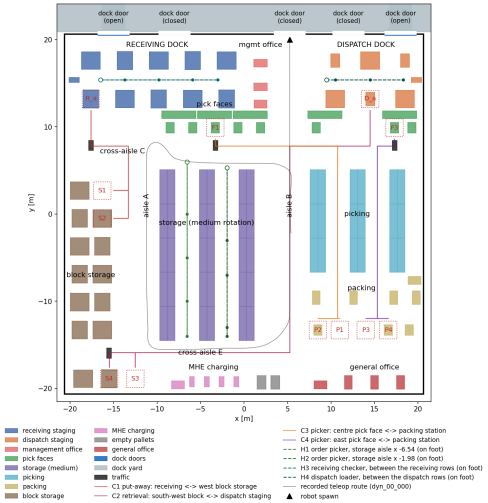
1. Realistic world
2. Experiment insights
3. `filter_mode:=motion`: why it does not work

1. Warehouse flow shapes in the design literature



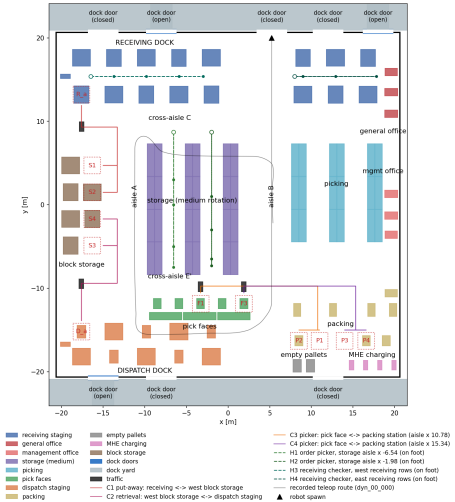
Three canonical layouts: I-shape (through flow), L-shape, U-shape. The realistic worlds implement the **I** and the **U**; the L was not generated.

1. U-shape layout: receiving and dispatch share the north wall



Filled rectangles: 126 objects by zone · coloured lines: 4 carrier routes · dashed green: 4 people · grey: recorded route of `dynamic_00_000`

1. I-shape layout: receiving north, dispatch south

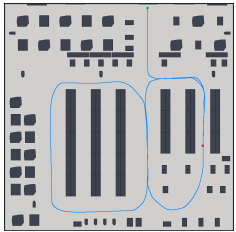


121 objects · same carriers and people · the grey recorded route crosses the racks here, so the I recordings use a new route

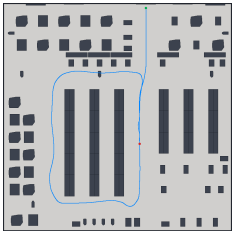
1. Four recordings: one per layout and twin, routes hand-driven

Recording	Layout	Traffic	Objects	Route [m]	Span [s]
realistic_000	U	no	125	171	126
realistic_001	I	no	119	133	97
realistic_002	U	yes	125	103	80
realistic_003	I	yes	119	158	117

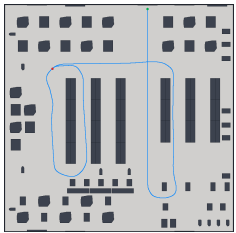
- Same object count within a layout (125 / 119): the twin differs only in whether the 4 carriers move
- Route column: the U pair differs by 68 m, so on U a static-vs-traffic difference mixes route with traffic



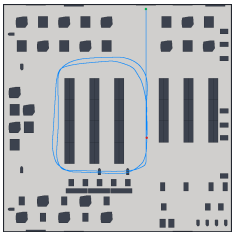
000 U, no traffic



002 U, traffic



001 I, no traffic

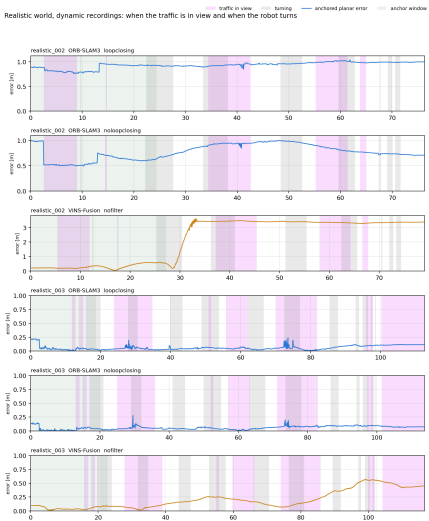


003 I, traffic

Source: final_report/tables/dataset_scenes.tex; blue trace = ground-truth route

1. When the traffic was actually in the camera: 37 % of tracked time

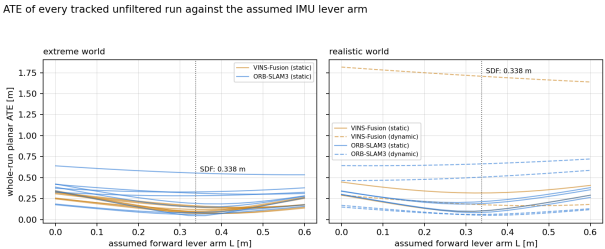
- Violet bands: a carrier or carried load within 20 m, within $\pm 45°$, line of sight clear, from the bag's pose stream
- Grey bands: robot turning ($>10\,°\,s^{-1}$)
- Median violet share over the tracked runs: 37 %
- People are not in the pose stream: no band for them



2. Scoring bug: every ATE curve bottoms out at the 0.338 m lever arm

- Ground truth = model origin; both estimators report the IMU, 0.338 m ahead in `model.sdf`
- Figure: each line is one tracked run re-scored with the ground truth moved forward by L; the minima sit at the dotted line
- Median $L^* = 0.34$ m, IQR 0.32–0.40, 30 runs, both estimators, both worlds

Static ATE [m]		$L = 0$	$L = 0.34$
extreme	VINS	0.32	**0.10**
extreme	ORB	0.37	**0.19**
realistic	VINS	0.37	**0.25**
realistic	ORB	0.32	**0.15**



ORB "drift while turning" in the realistic world: 8.7 mm s^{-1} at $L = 0$, 0.0 at $L = 0.34$ (8/8 runs).

Source: `exp3-insights/lever_arm_sweep.tex`, `lever_arm_argmin.tex`; fix applied in `vio_metrics.load_gt`, 22 Oct

2. Exp. 1: the EKF is the most accurate estimator on 1 bag of 5

Aligned planar ATE [m], bold = best per bag

Bag	EKF	VINS	ORB	ORB no LC
000	0.190	0.557	0.526	**0.167**
001	0.308	**0.127**	0.256	0.468
002	**0.256**	0.472	0.556	1.849
003	0.423	0.737	6.381	**0.313**
004	0.494	**0.275**	1.200	0.845

Clean RPE at 1 s [m]

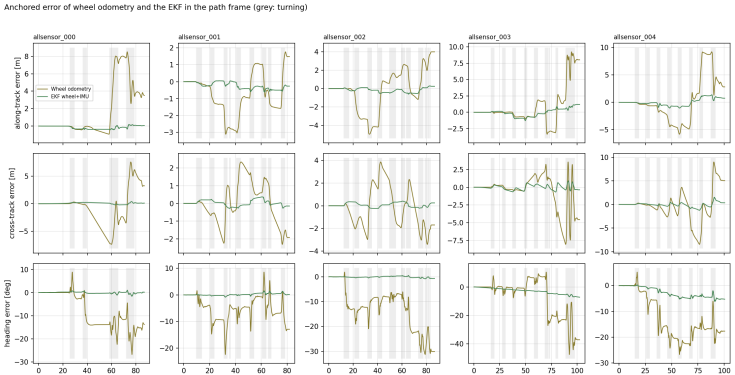
Bag	EKF	VINS	ORB	ORB no LC
000	0.088	0.034	0.027	**0.013**
001	0.092	0.019	0.018	**0.015**
002	0.076	0.094	**0.069**	0.085
003	0.105	0.112	0.177	**0.046**
004	0.140	**0.022**	0.035	0.024

Source: exp-estimator-baseline/proprio_metrics.csv, lever-corrected

- Top table, count the bold: EKF 1, VINS 2, ORB no LC 2
- EKF column is the only one under 0.5 m on all 5 rows
- Bottom table: EKF is never the best; visual estimators beat it on 5/5 bags
- Same EKF on the VSLAM noise file: mean ATE 0.334 $\rightarrow$ 1.135 m, loses to VINS 5/5 (not shown)
- VINS 000: 3-D ATE 0.557 m, planar 0.115 m; the difference is vertical drift the planar EKF cannot have

2. Exp. 1: wheel odometry's error is heading error made in the turns

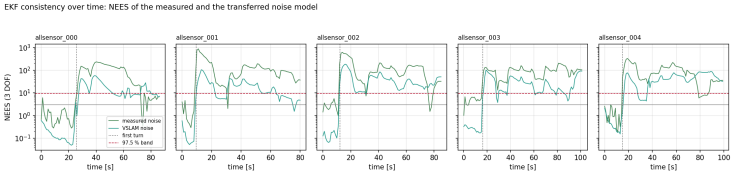
- Grey bands = turning
- Bottom row, brown: heading error steps *inside* the grey bands and holds between them, all 5 bags
- Top row: along-track (distance) error stays near zero until the heading error has integrated
- Scale error of the wheels: within 1 % on 5/5
- Green (EKF): flat in all three rows



Source: `exp1-insights/decomposition.tex`, `turning.tex`

2. Exp. 1: the EKF is consistent until the first turn, then never again

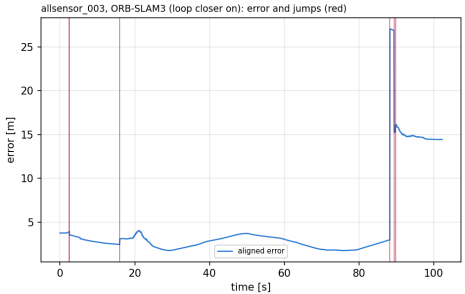
- Grey line: NEES = 3, the consistent value for 3 DOF
- Red dashed: 97.5 % band
- Dotted: first ground-truth heading change
- In all 5 panels the green curve crosses the band at the dotted line (within 0.6 s) and stays above it
- Teal (VSLAM noise file): less accurate but closer to the band



Source: `exp1-insights/nees_timing.tex`: leaves at 26.0 / 9.0 / 12.0 / 16.0 / 15.0 s; first turn at 25.6 / 9.6 / 12.2 / 16.4 / 15.0 s

2. Exp. 1: ORB-SLAM3's 6.4 m on `allsensor_003` is one loop-closure event

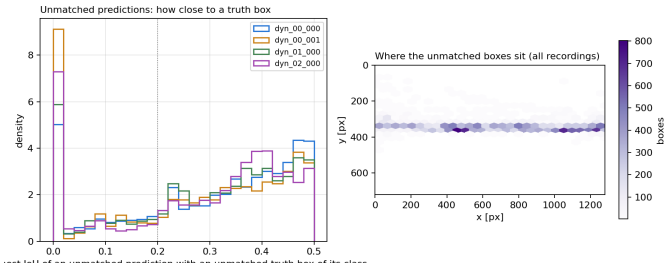
- Left: aligned error stays at 2–4 m for 88 s
- Red lines at 88–90 s: three jumps of 30.0, 11.6 and 2.5 m
- Right: the jump points (red dots) sit where the route closes its loop
- Tracking state OK on every frame; the no-loop-closure run scores 0.21 m



Source: `exp1-insights/visual.tex`

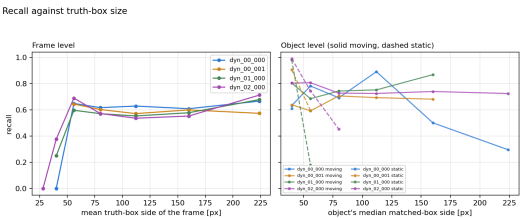
2. Exp. 2: 67–74 % of false positives are near misses on the right object

- Left histogram: best IoU of each unmatched prediction with an unmatched truth box of its class
- Mass right of the dotted 0.2 line = “right object, box too small”: 74 / 67 / 72 / 72 %
- Spike at 0 = nothing there: 6 / 15 / 9 / 10 %
- Right: the unmatched boxes sit in one horizontal band, the shelf height

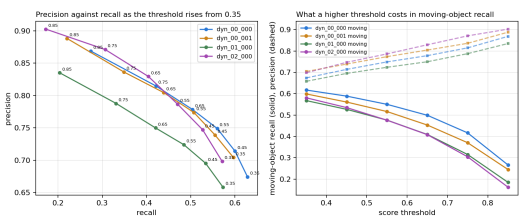


Source: exp2-insights/fp_anatomy.tex; 85–90 % of FPs are the box class (per_class.tex)

2. Exp. 2: neither size nor threshold explains the misses



Left panel: frame-level recall is flat at 0.57–0.66 for every box size from 48 px up; only the 32–48 px bin (121 of 62 282 boxes) is low.



Left panel: raising the threshold 0.35 → 0.55 moves precision 0.67 → 0.75 and recall 0.63 → 0.56 on `dyn_00_000`; the four curves trade one for one. Right: moving recall (solid) falls with the rest.

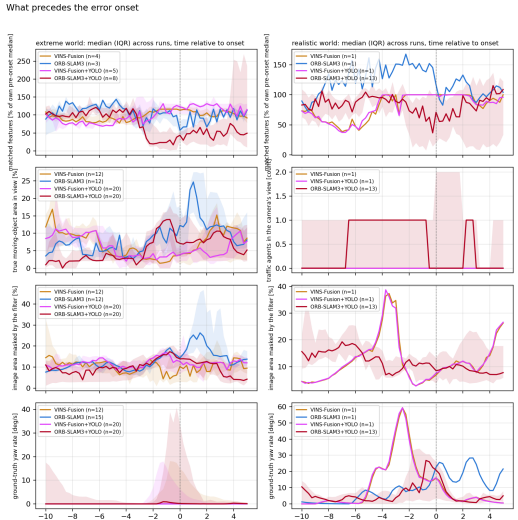
Source: `exp2-insights/recall_by_size.tex`, `threshold_sweep.tex`; moving recall is within 0.03 of overall on every bag

2. Exp. 3, extreme world: every dynamic run fails on the first straight

Onset of sustained error, 64 dynamic runs

Estimator	runs	none	turning
VINS-Fusion	12	0	4
ORB-SLAM3	12	0	0
VINS-Fusion+YOLO	20	0	3
ORB-SLAM3+YOLO	20	0	8

- 64/64 reach onset; 49 of them on a straight
- Moving objects are in view on 74–91 % of all frames, so "object in view" cannot separate anything
- Figure, top left: ORB+YOLO matched features (red) fall to ≈20 % of their own level 2.5 s before onset; VINS (orange) shows no drop



Source: `exp3-insights/onset_context_extreme.tex`; dotted line = onset

2. Exp. 3, realistic world: traffic in view does not change the drift

Drift rate while tracking, carrier in view vs. none

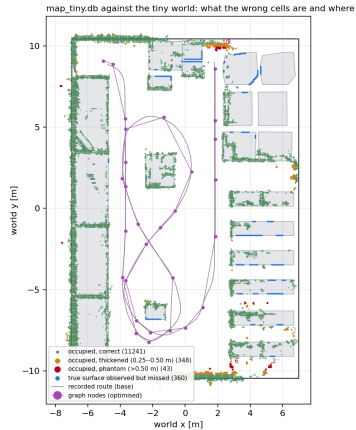
Estimator	n runs	in view [mm/s]	none	in>none runs	p
VINS-Fusion	1	2.0	5.7	0	—
ORB-SLAM3	4	−0.2	1.5	1	—
VINS-Fusion+YOLO	3	2.4	9.1	0	—
ORB-SLAM3+YOLO	10	42.0	8.2	7	0.08
matched features, ORB+YOLO	16	350	370	5	0.05

- Unfiltered rows: drift is not higher with a carrier in view (1 run of 5)
- Class-masked ORB: fewer matched features in view (11/16 runs) and faster drift (7/10): the mask removes features when a carrier passes
- Groups of 1–4 runs: a bound, not a measurement
- The one large step, VINS 002 3 m at 29–33 s, happens at the first turn with no carrier in view (slide 7, third panel)

Source: `exp3-insights/traffic_realistic_lever0p338.tex`

2. Exp. 4: the wrong cells are thick surfaces, not phantoms

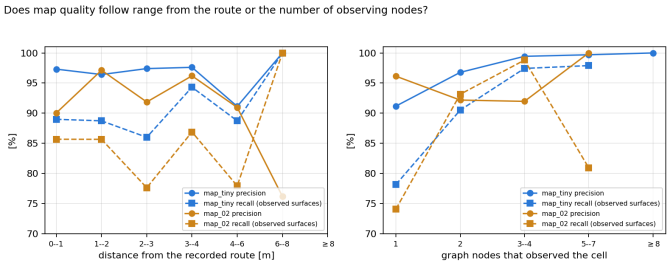
- Legend counts: green 11 241 correct, orange 348 thickened (0.25–0.50 m from a true surface), red 43 phantom (>0.50 m)
- The 43 red cells form 22 clusters; the largest (labelled 10, 6, 2) sit at the north and south walls, none in an aisle the route uses
- Blue 360 = true surface observed but missed; they lie on the shelf faces across the east-side gaps, seen edge-on from the aisle



Source: exp4-insights/cell_classes.tex; precision 96.6 % = 11 241 / 11 632

2. Exp. 4: surfaces are missed when seen once or edge-on, not when far

- Left panel: precision and recall show no trend from 0 to 6 m from the route
- Right panel: recall 78 % for cells seen by 1 node → 91 % at 2 → 97 % at 3–4; precision 91 → 97 → 99 %
- Incidence angle (table, not shown): recall 95 % at 0–30°, 92 % at 30–60°, 76 % at 60–90°



Source: `exp4-insights/bins_distance.tex`, `bins_views.tex`, `bins_angle.tex`; `map_tiny` and the second tiny map `map_02`

2. Exp. 4: the databases nobody had registered

File	Date	Nodes	What the contents say	Evidence
`map_tiny.db`	9 Sep	—	built from `dataset_map_tiny`	node poses match its ground truth, median 0.05 m
`map_02.db`	21 Sep	—	a second tiny map, other route	grid fits the tiny world; drivability 7.6 % blocked
`map_03.db`	21 Sep	18	the large-warehouse attempt	76 m of poses climbing to 9.8 m; no world fits
`map_01.db`, `map_02_vins.db`	21 Sep	2	aborted starts	two nodes each

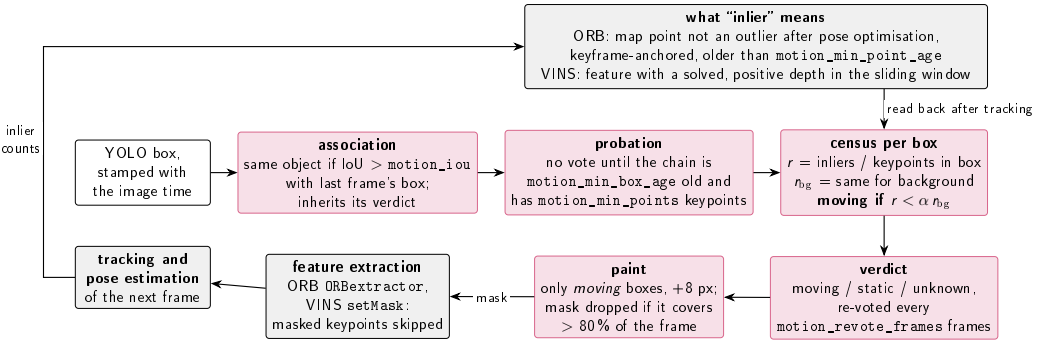
The raw write-up says the scored map "matches no recorded dataset" and the large attempt "left no artefact". Both are wrong.

Source: `exp4-insights/databases.tex`

2. What changes in Chapter 4

Was	Is	Slide
Static ATE 0.3–0.6 m; ORB error grows at turns	0.1–0.25 m; the growth was the lever arm	8
EKF beats VSLAM	EKF best on 1/5 by ATE; worst on 5/5 by RPE	9
EKF consistency as whole-run ANEES	consistent until the first turn	11
ORB 6.4 m unexplained	one loop closure at 88 s	12
Detector misses moving objects more	one localisation deficit on `box`	13–14
Traffic degrades tracking	no measurable effect	16
`map_tiny` unmatched; large map lost	`dataset_map_tiny`; `map_03.db`	19

3. How the motion gate works (as built)



Defaults: $\alpha = 0.5$, box age 30 frames, point age 30 frames (ORB), re-vote every 15 frames; the evaluation's best operating point was $\alpha = 0.25$. Only boxes judged *moving* lose their features. 56 % of box-frames reached a vote; 13–16 % ended up masked.

Source: `filter_motion.tex` chapter 3

3. `filter_mode:=motion`: the gate cannot tell moving from stationary

ORB-SLAM3 census, `dynamic_00_000`, ground-truth labelled box-frames

ratio $\leq$	moving flagged	static flagged	J	AUC
0.25	0.820	0.760	+0.061	0.528
0.50	0.854	0.809	+0.045	
0.75	0.875	0.840	+0.035	
1.00	0.888	0.868	+0.020	

3 307 moving and 1 044 static box-frames; AUC 0.5 = coin toss. Older boxes score worse (AUC 0.40 at age 20).

VINS-Fusion, three census signals

Signal	recall	false alarm
solved depth	0.25	0.23
reprojection residual	**inverted**: movers score lower	
residual availability	0.56	0.41

- Every row flags stationary objects almost as often as moving ones
- Status: implemented in both pipelines, disabled by default; Exp. 3 reported on class mode

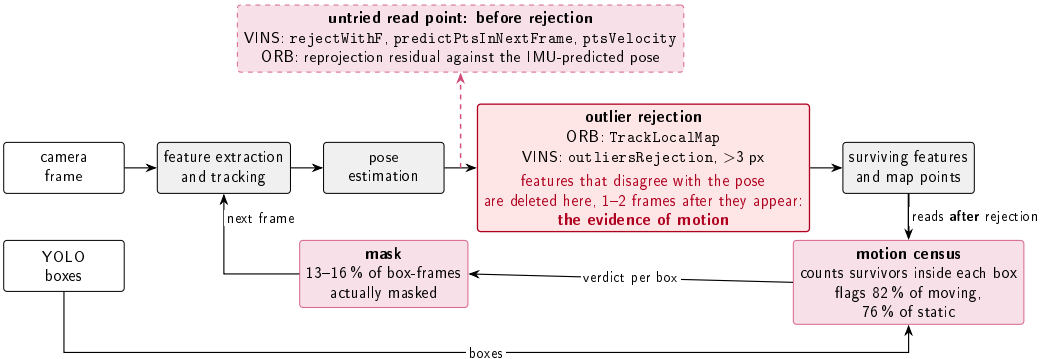
Source: `exp3-motion-gate/motion_gate_summary.csv`, `filter_motion.tex` table 4.2

3. Why: the census reads state the estimator has already cleaned

- ORB-SLAM3 counts map points that survived `TrackLocalMap`; points inconsistent with the optimised pose were marked outliers first. What is left inside a mover's box is whatever happened to fit, hence recall $\approx$ false alarm in every row of the previous slide
- VINS-Fusion `outliersRejection` removes any feature above 3 px reprojection error every frame; a mover's disagreeing features are gone within a frame or two, so the residual census measures survivors. That is why the residual signal came out **inverted**
- Test that confirms it: lowering `min_dist` 30 $\rightarrow$ 15 px tripled the features per box (coverage 19.6 $\rightarrow$ 55.8 %), and the depth signal vanished: 94.7 % of features inside moving objects held solved depths, 0 of 83 voted rows fell below threshold. More data, no signal
- A moving point triangulates to a wrong but valid depth; no depth test separates it
- Architectural, not tuning: no threshold on these quantities can repair it

Source: `filter_motion.tex`, chapters 4–5

3. Where the gate reads, and where the evidence is lost



Grey: the estimator. Purple: what was built. Red: where the signal is destroyed. Dashed: where a motion signal can still exist.

3. What could work: signals taken *before* rejection (untried, no data yet)

- VINS: `rejectWithF()`, RANSAC fundamental-matrix check, written but commented out; its outliers violate epipolar geometry by definition
- VINS: `predictPtsInNextFrame()`, implemented, never called: predicted-vs-tracked displacement per feature
- VINS: `ptsVelocity`, per-point image velocity already computed every frame; ego-motion is a smooth field, a mover a local anomaly
- ORB-SLAM3: reprojection residual of each keypoint against the IMU-predicted pose in `Tracking`, before outlier marking
- Rule: nothing is quoted until `analyse_motion_gate.py` has scored it against ground truth with the sample size stated (19 rows looked good, 1044 did not)

Next steps

1. Rewrite Chapter 4 prose from the insight documents (tables and figures already regenerated)
2. Quote planar ATE beside 3-D for the visual estimators
3. Re-run `realv3` on the 21 Oct extrinsics; measure the tracker lever arm
4. Log per-box detector truth; re-score at a lower IoU
5. Place `map_03.db`; score `map_02.db` with provenance stated